Atlas 210X — VFO Nano add-on wiring

YT3CT (ex YU5KBM) · Nano ATmega328P (Old Bootloader) · sketch atlas_vfo_fixed.ino · 2026-10 · button / relay part = proposal

1. LCD RG1602A (16×2) — twisted wiring, 12 wires

Twisted group	Nano board pad	LCD pin
orange / black	orange: D8 → 470 Ω	6 E
	black: GND	1
blue / black	blue: D7 → 470 Ω	4 RS
	black: GND	1
red / black / yellow / grey	red: +5 V	2 VDD
	black: GND	1
	yellow: contrast pot wiper	3 V0
	grey: 10 Ω pad (backlight)	15 A
brown / white	brown: D9 → 470 Ω	11 D4
	white: D10 → 470 Ω	12 D5
purple / green	green: D11 → 470 Ω	13 D6
	purple: D12 → 470 Ω	14 D7

- **Nano board:** pads unchanged; all 3 black wires on the one GND pad. 470 Ω in line within ~2 cm of the pads (6 signal wires only).
- **At the LCD:** existing jumper 16–5–1 stays; 100 nF + 10 µF across pins 2–1. Pins 7–10 open.
- Different twist per group, < 1 cm untwisted at the ends. Groups side by side along the chassis.

2. TTL programming leads (no USB needed)

USB-TTL cable (Prolific)	Nano
Green (TX)	D0 / RX
White (RX)	D1 / TX
Black (GND)	GND
Red (+5 V)	**NOT connected**
—	RST lead (+ optional push button to GND)

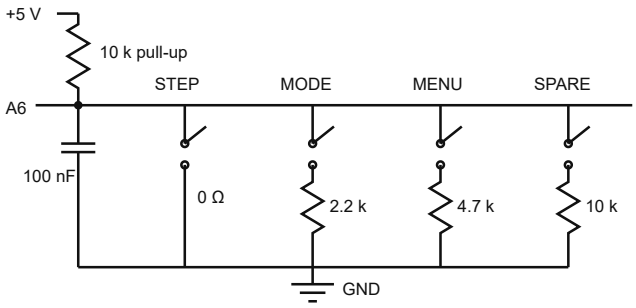
- If no sync: swap green / white (colours differ between cables).
- **Upload:** start upload, then short RST to GND for ½ s. Power cycling only works if the radio stays off until the Nano POW LED is fully out (TTL line back-feeds the Nano).
- Same procedure for the VariBeam Nano.
- Never plug the Nano USB in while the TTL cable is connected.

Free Nano pins & plan proposal

Pin	Now	Plan
D0 / D1	Serial	TTL leads
D4	free	CW sense (+CW)
D5	free	ATT switch sense
D6	STEP button	PRE switch sense (STEP → A6)
D13	free (LED)	AGC fast switch sense
A6	free (analog only)	STEP / MODE / MENU ladder
A7	free (analog only)	spare

Used: D2/D3 encoder, D7–D12 LCD, A0/A1 S-meter, A2 TX, A3 band, A4/A5 I²C Si5351.
Sense inputs: 22 k from the switched +12/13 V line, 10 k to GND at the pin (≈ 4 V). Never the raw +13 V on a Nano pin.

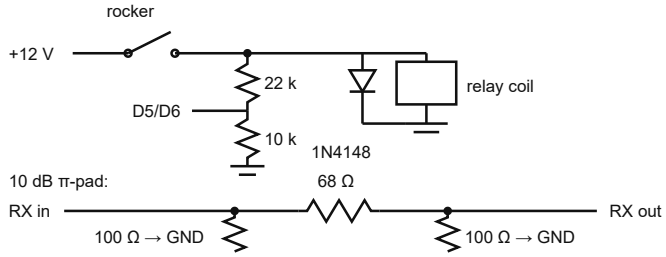
3. Buttons on A6 (resistor ladder) proposal



Button	R to GND	ADC	Accept if
STEP	0 Ω (direct)	0	< 90
MODE LSB/USB	2.2 k	184	90–255
MENU	4.7 k	327	255–420
SPARE	10 k	512	420–770
none	—	1023	> 770

A6/A7 have **no internal pull-up** → the external 10 k is required. One button at a time; SPARE optional. Resistors 1 % or 5 % both fine.

4. ATT / PRE relays (rockers) — RX only



- π-pad 100 / 68 / 100 Ω ≈ 10 dB (exact: 96 / 71 / 96 Ω). Relay bypasses or inserts it. PRE relay wired the same way (switch, coil, sense to D6).
- **Only in the receive line after the antenna relay — never in the TX path.** Short leads / thin coax.
- Better: one ON-OFF-ON rocker for ATT – OFF – PRE (both can never be on). Diode cathode (band) to + side.

5. AGC fast/slow — rocker switch, sense D13

- 2N7000 at PC-300D: gate via 100 Ω from the rocker (+12 V), 100 k gate–GND, source GND, drain via R_{fast} (~150–330 k, to try) to the AGC hold node on PC-300D.
- Parts on PC-300D to be identified by value first (hold cap / 1 M discharge R).

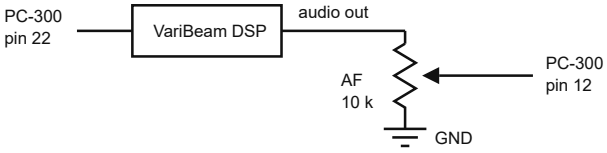
6. MODE (button) · CW (sense D4)

Software only: BFO moved to the other side of the crystal filter (5.520.000 ↔ ~5.523.3 MHz, to be set) — LO calculation follows automatically.

7. Front panel gain controls — 2 concentric dual pots, fixes for the new values

Pot	Function	Orig.	Fitted	Fix
1	AF gain (R4)	10 k	10 k	none
	RF gain (R6)	10 k	5 k	R5 4.7 k → 2.2 k, R7 470 → 220 Ω
2	MIC gain (R12)	10 k	20 k	20 k across outer lugs
	ALC (R11)	10 k	100 k	11 k across outer lugs

- RF gain: GND–R5–pot–R7–+8.5 V, wiper to PC-100 pin 14; wiper range ~2.6–8.2 V again. Measure R5 first (if already ~2.4 k, only R7).
- Resistor across the outer lugs = 10 k load as original, wiper ratio unchanged. 100 k ALC = harder, slower ALC → first suspect for low modulation.



- AGC / S-meter pickoff is before pin 22. Shielded audio leads, shield grounded at one end.
- R13 10 k (mic gain top) goes to the WHT/BLK **+CW** line (S3 CW / PC-100 pin 18), 0 V in SSB. Not PC-100 pin 21 (+13 V on every TX).